

Benchmark Datasets for Hate Speech Detection Algorithms

Sid Diamond

April 14, 2026

Abstract

Machine learning benchmarks shape the development and evaluation of modern decision-making systems, including hate speech detection algorithms. Focusing on three hate speech detection papers ([1], [2], [3]) and the datasets they use, this paper makes two claims. First, these benchmarks contain biases that restrict equality of opportunity by influencing outcomes on grounds other than merit. Second, using Fishkin’s theory of bottlenecks [4], as adapted by Jain et al. for algorithmic fairness [5], I argue that hate speech benchmarks act as *severe bottlenecks*: decision points that restrict access to spaces governed by these systems and thereby also restrict equality of opportunity. This is problematic because platforms moderated by such systems are now central sites of public discourse and political influence. I further argue that, while bias and bottleneck severity are distinct, reducing bottleneck severity can mitigate both and improve equality of opportunity for those seeking to participate in these spaces.

1 Introduction

Given the growing importance of decision-making algorithms, it is crucial to establish scientific, comprehensive, and fair systems to benchmark their development. For a powerful family of algorithms called supervised machine learning algorithms, these systems are referred to as benchmark tests, which quantitatively assess model performance at important stages of their development.

Supervised machine learning algorithms are used in many automated decision-making processes, from disease prediction to financial advice [6],[7]. For example, AI adoption among UK financial advisers rose from 21% to 37% last year alone [7]. This growing use has increased the importance of benchmark tests, which are used to evaluate and optimise these algorithms for their specific tasks.

Here, I examine benchmark datasets used to train supervised machine learning systems for automated hate speech detection. Online hate speech is widespread: Gagliardone et al. (2015) reported in the European Union, 80% of people encountered it, and 40% felt attacked or threatened on social networking sites [8]. Social media platforms now serve as major spaces of public discourse and political influence [9]. Exclusion from these spaces, whether through intimidation or potentially unfair algorithmic bans, can restrict opportunities to participate in public debate.

For hate speech detection systems, training datasets determine which users and content are allowed to remain in the spaces these systems moderate [10]. I argue that hate speech detection benchmark datasets contain two major, related flaws that restrict equality of opportunity in these spaces.

First, because there is no clear definition of hate speech, hate speech detection datasets inevitably contain bias [3], which benchmark datasets amplify [11]. This bias restricts equality of opportunity by shaping access to these spaces on grounds other than merit or desert [5].

Second, drawing on Jain et al.’s [5] adaptation of Fishkin’s theory of *opportunity bottlenecks* [4], I argue that these benchmarks constitute *severe bottlenecks*. A bottleneck is a decision point that restricts access to opportunities, and its severity depends on how pervasive it is within a system and how strict the consequences of its judgments are. In hate speech detection, the relevant decision point is which content, and therefore which users, are allowed to and feel safe enough to participate in spaces moderated by hate speech detection algorithms. Fishkin argues that severe bottlenecks necessarily limit equality of opportunity in the spaces where they operate.

Although these two problems are distinct, I argue that they are connected because both can be mitigated through a single solution. Reducing bottleneck severity can improve equality of opportunity in spaces moderated by these systems both directly, by making access less restrictive, and indirectly, by limiting the impact of dataset bias.

2 What is a Benchmark Test?

A benchmark test can be understood, at a basic level, as a dataset constructed from a collection of input–output pairs [12]. In sentiment analysis the inputs might be tweets or movie reviews, while the outputs are labels assigned by humans or algorithms that indicate sentiment, such as whether a text is positive, negative, hateful, or not [13, 1, 2].

There is, however, no single settled definition of a benchmark test. Benchmarks serve a range of purposes, and datasets originally designed for one task are often repurposed for others [14]. In machine learning engineering a dataset typically becomes a benchmark through repeated use in influential peer-reviewed research. An example best illustrates this.

Consider the transformer architecture, the “T” in GPT (Generative Pre-trained Transformer). The transformer was originally evaluated on the WMT 2014 English-to-German and English-to-French translation datasets, containing 4.5 million and 36 million sentence pairs respectively [15].

The transformer was originally developed to improve machine translation, and evaluation on established translation datasets suited that purpose. However, given the architecture’s enormous influence, the WMT datasets gained further legitimacy through their association with its success. They now function as broader indicators of quality for general-purpose models whose uses extend far beyond translation. Now, strong performance on WMT-style tasks can be treated as evidence of overall algorithmic quality, even though these tasks capture only a narrow aspect of performance.

This example helps explain why datasets originally designed for one task are often repurposed for others. Koch et al. make this point explicitly [14]:

Claim 1 (Few Datasets). *“Machine learning research communities are increasingly concentrating evaluation on smaller subsets of benchmark [tests], and frequently reuse datasets originally developed for other tasks.”*

3 Data

3.1 Literature Selection

A primary evaluation of specific benchmark datasets is neither necessary nor within the scope of this research. It would require examining carefully selected subsets from datasets containing thousands of data points. Instead, I rely on secondary literature that analyses the datasets relevant to my argument. The studies on hate speech detection benchmark datasets considered in this work are:

1. *Automated Hate Speech Detection and the Problem of Offensive Language* [1].
2. *Hateful Symbols or Hateful People? Predictive Features for Hate Speech Detection on Twitter* [2].
3. *The Risk of Racial Bias in Hate Speech Detection* [3].

Hate speech detection datasets are the object of study because improving the identification of online hate speech is ethically important: hate speech classifications shape who is excluded from social media platforms, whether through intimidation or unfair flagging and bans. As Bousquet (2025) argues, exclusion from these platforms can limit opportunities to participate in public debate and to contest how one’s views or identity are represented [9]. This is

especially significant for marginalised communities, for whom social media can provide an alternative space to challenge dominant public discourses [16]. In some cases, exclusion may also carry professional or economic costs, given the role social media plays in areas such as journalism [9]. For instance, 94% of journalists in the U.S. use social media for their work, relying most heavily on Twitter and Facebook [17].

Large-scale hate speech detection systems help determine what content remains online, who is banned for what these systems classify as hate speech, and who feels too threatened to participate because harmful content is left unremoved. In large-scale moderation, supervised machine learning is commonly used to classify content and prioritise posts for review. As a result, the way benchmark tests are designed and used shapes which posts moderators review [10]. So, if hate speech detection algorithms are of ethical significance given that they decide who gets what access to participate in significant places of public discourse, then as are the hate speech benchmark datasets they are trained on.

I believe papers 1, 2, and 3 can be taken as representative of the datasets used to train automated hate speech detection algorithms. Regarding these datasets as such is not a unique opinion. In introducing the FDCL18 hate speech detection dataset [18], Founta et al. situate their dataset among five "related datasets", including the three examined here, alongside those of Chatzakou et al. and Golbeck et al. [19, 20]. Like Founta et al., I take this to suggest that the datasets considered in this chapter are not only benchmark datasets, but also reasonably representative of other datasets in the field of hate speech detection. The three chosen papers were selected using two criteria:

Criteria 1. *Influence in the field.*

Criteria 2. *Sufficiently explicit data collection methods.*

Criterion (1) is used to identify whether a dataset belongs to the small but influential subset described in Claim (1). Analysing these widely used datasets is therefore disproportionately informative about the field. I approximate influence by citation count. Whilst not exact, this is (mathematically at least) reasonably well-grounded given the power-law structure of citation networks [21]. The least cited paper examined here, paper 3, has 1,325 citations, which I take to indicate significant influence in hate speech detection research.

Criterion (2) is used because this work is an analysis of datasets. The selected papers must therefore provide enough detail about how the data were collected and labelled to permit critical evaluation of any bias introduced in those processes. I include not only original dataset papers, but also secondary studies that clearly describe and justify the datasets they analyse.

3.2 The Papers Studied

3.2.1 Automated Hate Speech Detection and the Problem of Offensive Language

Paper 1, has 4,007 citations. Davidson et al. train a hate speech detection model on 24,802 tweets sampled using *Hatebase.org*, a collaborative repository of multilingual hate speech, and labelled by CrowdFlower workers. CrowdFlower (now Appen) was a crowdsourcing platform used to distribute many small online tasks, often for low pay. Paper 1 does not report the specific characteristics of its CrowdFlower workers.

This limited transparency about annotator selection, an omission not unique to paper 1, is a significant ethical flaw in hate speech detection datasets. It leaves unresolved questions about who the annotators are, how they are chosen, and how their selection shapes decisions about exclusion from these important spaces of public discourse.

The tweets were labelled as hate speech, offensive but not hate speech, or neither, with each tweet coded by at least three annotators and an inter-annotator agreement of 92% [1]. Following Sap et al. [3], I refer to this as the DWMW17 dataset. Given the stated sample size and annotation procedure, I take criterion 2 to be satisfied.

Using the DWMW17 dataset, Davidson et al. show that racist and homophobic tweets are more likely to be classified as hate speech, while sexist tweets are more often classified as merely offensive [1]. Evidence of what this implies can be seen in 23% of women reporting being comfortable expressing political opinions online, compared with almost 40% of men in a 2024 UK survey [22]. The key point is that Davidson et al. suggest this arises because social biases in annotation can be embedded in datasets and reproduced in downstream systems, restricting equality of opportunity for those subject to decisions made by these systems. The dataset’s wider use in influential research supports classifying it as a benchmark dataset [3, 23, 18].

3.2.2 Hateful Symbols or Hateful People? Predictive Features for Hate Speech Detection on Twitter

Paper 2 has 2,552 citations. Waseem and Hovy develop criteria for identifying hate speech, informed by critical race theory, and use them to annotate a publicly released dataset of over 16,000 tweets. The paper reports the sample size, collection window, and annotators, so I take criterion 2 to be satisfied.

Compared with many hate speech datasets, this one has a more explicit sociological basis. However it was labelled by only three annotators, whose individual biases are unavoidable and unlikely to reflect the broader communities affected by systems trained on the dataset. The key point is that biases introduced by a small annotator pool are reproduced and amplified as the dataset is reused in later influential work. This highlights the importance

of sufficiently large and diverse annotator pools in datasets that shape access to moderated online spaces. Given its citation count and wider use in other influential work (e.g. in FDCL18 [18]), I treat this as a benchmark dataset.

3.2.3 The Risk of Racial Bias in Hate Speech Detection

Paper 3 has 1,325 citations. Sap et al. examine how dialect differences can produce racial bias in automatic hate speech detection by analysing two benchmark datasets: FDCL18 [18], which contains 80,000 tweets labelled by five to twenty crowd-sourced workers as hateful, abusive, spam, or none, and DWMW17 [1]. Like DWMW17, FDCL18 omits important details about who its annotators were beyond describing them as crowd-sourced, and so shares the same lack of labeller transparency identified in paper 1. Still, because both datasets are described in enough detail to permit at least some critical evaluation, I take criterion 2 to be satisfied here too.

Sap et al. test the relationship between dialect and hate speech classification using a model that estimates whether tweets are written in African American English or white-aligned English [3, 24], allowing them to assess systematic differences in outcomes. They find that models trained on these datasets reproduce racial dialect bias, with African American English tweets about twice as likely to be labelled offensive. They attribute this in part to the difficulty of identifying reclaimed language: terms reclaimed within a community but still offensive when used by outsiders [25]. The key point this highlights is that fixed label categories often fail to capture the contextual complexity of hate speech, including who is speaking and in what context.

4 Opportunity Bottlenecks

4.1 Opportunity Ethics

Any attempt to make algorithmic decision-making fairer must begin by defining the fairness we wish to implement. Following Jain et al., I use a practical definition of fairness grounded in the principle that *all persons have equal moral worth and should therefore be treated equally* [5]. The question is: What, exactly, should be equalised in algorithmic decision-making?

There is no consensus on this issue. Competing proposals include demographic parity, which equalises acceptance rates [26]; equalised odds, which equalises the distribution of outcomes across groups [27]; and calibration, which equalises the meaning or reliability of predictive scores [28]. Despite these disagreements, these approaches share a commitment to equality of opportunity [29]. Again following Jain et al., I contend that *"formal views of equal opportunity often contend that all decision subjects should have a chance at a positive outcome and that their likelihood of receiving that outcome accords with their merit or desert."* [5].

Papers 1, 2, and 3 each identify ways in which the datasets they discuss ([1], [2], and [18]) can restrict equality of opportunity when it is thought of in this way. Paper 1 suggests that social biases among annotators can lead sexist abuse being flagged as hate speech less often than other hateful content, placing women at a comparative disadvantage in online participation. Paper 2 shows how a dataset labelled by a small number of annotators may encode their biases and create unequal barriers to participation. Paper 3 shows that failure to identify reclaimed speech can lead African American English tweets to be labelled offensive at substantially higher rates. In all three cases, access to the opportunities associated with spaces governed by algorithms trained on these datasets is distributed unequally on grounds other than merit or desert.

I now argue that equality of opportunity is reduced not only by biases within these datasets, but also by their pervasiveness across the online spaces they moderate and the strictness of the outcomes they inform (i.e. what gets flagged as hate speech). To show this, I turn to Fishkin’s theory of *opportunity bottlenecks* [4].

4.2 Fishkin Bottlenecks

Individuals of equal merit and desert should have equal opportunity to participate in spaces of public discourse [9, 16], and thereby to access the public and professional opportunities linked to such participation [17]. Regional and cultural dialect differences should not affect that access. Reclaimed hate speech shows why: terms that are hateful in some contexts may not be hateful when used by the community reclaiming them [25]. This is not merely hypothetical. As paper 3 shows, tweets written in African American English are disproportionately labelled as hate speech because systems trained on benchmark datasets such as DWMW17 and FDCL18 fail to account properly for reclaimed speech. Since these datasets are representative of the field, this is a real example of unequal opportunity generated on grounds other than merit or desert.

To introduce Fishkin’s theory of bottlenecks, I turn to a thought experiment adapted from Fishkin’s [4] adaptation of Williams’s warrior example [30]. I recast it as in a dialect-centred case to show how bottlenecks can restrict opportunity even before dataset bias is considered.

Imagine a world in which children everywhere have equal opportunities to learn the dialect they will use throughout life. In this (admittedly boring) world, dialect is taught identically in all classrooms. Suppose that a single supervised machine learning system, whose outputs are shaped by a small number of training datasets, flags speakers for “improper dialect,” and that being flagged significantly worsens one’s quality of life by restricting access to public discourse and economic opportunity. All citizens begin life with equal opportunity regarding who is flagged by this algorithm.

There will, however, still be differences in dialect between individuals. Some will be naturally better suited to the aspects of this dialect treated as “proper,” just as some are better suited to the particular ways mathematical ability is tested. These differences arise even before

we introduce dialect-related bias in the datasets used to train such systems. Even so, those disadvantaged by failing to excel in this dialect could still object that the bottleneck through which they must pass to succeed is simply too narrow. In Fishkin’s terms, a bottleneck is a “*mechanism through which opportunities are created, distributed, or destroyed*” [4].

Each bottleneck limits access to opportunities with some degree of severity and legitimacy. Severity is the degree to which it constrains opportunity [5], and depends on two factors: *pervasiveness* and *strictness*. Pervasiveness is the range of people and opportunities affected; strictness concerns where the bottleneck lies on the spectrum between absolute bar and no preference in outcomes [5, 4].

Fishkin argues that severe bottlenecks constrain equality of opportunity regardless of their legitimacy [4]. I therefore set legitimacy aside and focus on how reducing bottleneck severity can increase equality of opportunity.

In this thought experiment, the algorithm is pervasive because everyone is subject to it, and strict because its consequences are severe. It is therefore a severe bottleneck. Since the small number datasets training the algorithm determine its behaviour, these too function as severe bottlenecks. This, however, remains only an analysis of the thought experiment. The real question is: To what extent are actual hate speech detection benchmark datasets severe bottlenecks?

4.3 Hate Speech Detection Datasets are Severe Bottlenecks

To show that hate speech detection benchmark datasets are severe bottlenecks, two conditions must be met:

condition 1. *Hate speech detection datasets significantly affect the outputs of the algorithms they train.*

condition 2. *Hate speech detection algorithms are severe bottlenecks in the spaces they moderate.*

Condition 1 follows from the nature of supervised machine learning. Hate speech detection systems map real data to the same labels used to classify training data [31, 12]; their outputs are shaped by the examples and classifications on which they are trained [32, 31]. As Hammoudeh and Lowd put it, “*all model decisions are rooted in the training data*” [32]. I therefore take this condition to be met.

To assess condition 2, I focus on a particular online space and its moderation practices. Following section 3.1, I treat papers 1, 2, and 3 as broadly representative of the field in both dataset content and accounts of how hate speech detection systems operate. On this basis, Twitter between 2016 and 2018 is a reasonable case study.

Twitter’s public transparency materials indicate that content moderation at the time relied on supervised machine learning trained on platform data, including content previously reviewed by moderators, random samples, and other platform-derived samples [33]. However, these materials do not identify the datasets used to train hateful-conduct models, report the size or composition of the training corpus, specify the model architectures, or make the datasets publicly available.

The available evidence suggests that Twitter’s systems were trained on internally labelled Twitter data rather than on the benchmark datasets used in academic hate speech detection research. At most, this internal data and modelling approach appear broadly similar to those used in papers 1, 2, and 3. Given this opacity, academic work can do little more than make informed attempts to reconstruct such models, as in Tonneau et al., who show that training data significantly influence which users and content are permitted to remain in the spaces these systems moderate [10].

This weakens any demonstration of condition 2 and underscores the need for greater transparency about platform training data. As the *Digital Regulation Cooperation Forum* notes, effective algorithmic auditing is often hindered by “*lack of access to systems*”, while meaningful scrutiny may require access to “*the data on which the model has been trained*” [34].

Still, this does not show that Twitter’s moderation algorithms were not severe bottlenecks. It means only that condition 2 cannot be demonstrated rigorously with the available evidence. Even so, given the broad similarity between Twitter’s apparent approach and the models studied in papers 1, 2, and 3, together with the findings of Tonneau et al. and the dialect thought experiment above, it is reasonable to conjecture that Twitter’s hate speech detection systems between 2016 and 2018 were severe bottlenecks on equality of opportunity for those seeking to participate in that space.

I therefore conjecture that, if more information were available about the datasets and algorithms used by Twitter between 2016 and 2018, or by similar online spaces, these systems would be found to satisfy condition 2. If so, actual hate speech detection benchmark datasets would be severe bottlenecks on equality of opportunity.

5 Dataset Severity and Bias

In Section 3, I argue that papers 1, 2, and 3 each identify distinct forms of bias, and show that in each case these affect the chance of a positive outcome on grounds other than merit or desert, thereby restricting equality of opportunity.

In Section 4, following Fishkin [4], I outline how severe bottlenecks constrain equality of opportunity. I then argue hate speech detection benchmark datasets (e.g. [1], [2], [18]) are severe bottlenecks as they limit equal access to important spaces of online discourse [9].

As shown in the thought experiment in Section 4.2, severe bottlenecks restrict equality of opportunity even in the absence of bias. Similarly, dataset bias can persist regardless of the severity of the bottleneck. However, reducing bottleneck severity can improve equality of opportunity in both cases. To show this, I introduce a concept linking bottleneck severity and dataset bias: *algorithmic monoculture* [35].

Algorithmic monoculture describes systems that repeatedly place the same groups on the losing side of important decisions [35]. In supervised machine learning, this can occur when many systems rely on the same small set of datasets and so reproduce similar exclusions. This is precisely the risk identified by Koch et al. in claim 1. Bias is central to this problem: when widely reused training data contain implicit social bias, those biases can shape repeated decisions across platforms. For example, the datasets used in papers 1 and 3 can respectively disadvantage women through under-flagged sexism and speakers of African American English through over-flagged reclaimed speech. This repeated disadvantage is a central concern of algorithmic monoculture.

Jain et al. show that algorithmic monoculture is also problem of bottleneck severity: *"The concern about algorithmic monoculture is a concern about severe bottlenecks"* [5]. This is because *"outcome homogenization constitutes a pervasive bottleneck for those that repeatedly receive undesirable decisions because they may find themselves locked out of many paths in the opportunity structure"* [5]. Algorithmic monoculture therefore links bias and bottleneck severity: the more pervasive the bottleneck, the more widely its underlying biases shape decision outcomes.

It follows that reducing bottleneck severity also reduces the impact of these biases on equality of opportunity in spaces moderated by systems trained on these benchmark datasets. In other words, the effect of bias depends on bottleneck severity: if the bottleneck is severe, even slight biases can be amplified and significantly affect outcomes. If a dataset contains significant bias but does not create a severe bottleneck, that bias is less likely to substantially affect equality of opportunity.

So, how can we reduce the severity of the bottleneck created by hate speech detection benchmark datasets? This paper's contribution to that question is to show how Jain et al.'s account of algorithmic pluralism applies specifically to hate speech detection benchmark datasets. I argue that these benchmarks not only contain bias but also function as severe bottlenecks governing access to online discourse. The severity of these bottlenecks in turn amplifies those biases, producing a form of algorithmic monoculture [35]. When viewed this way, the procedure outlined by Jain et al. in Section 4.2, *"Pluralism in Algorithmic Decision-Making,"* offers a useful starting point for thinking about how such bottlenecks might be reduced in this domain [5].

6 Conclusion

In conclusion, by drawing on three influential studies (1, 2, 3), this work shows that hate speech detection benchmark datasets contain biases that can restrict equality of opportunity by shaping outcomes on grounds other than merit or desert. Using Fishkin’s theory of bottlenecks [4], as adapted by Jain et al. [5], I further argue that they function as severe bottlenecks governing access to online discourse and related opportunities [9]. Although bias and bottleneck severity are distinct, both can be mitigated by reducing bottleneck severity, thereby improving equality of opportunity.

References

- [1] Thomas Davidson, Dana Warmusley, Michael Macy, and Ingmar Weber. Automated hate speech detection and the problem of offensive language. In *Proceedings of the 11th International AAAI Conference on Web and Social Media*, ICWSM ’17, pages 512–515, 2017.
- [2] Zeerak Waseem and Dirk Hovy. Hateful symbols or hateful people? predictive features for hate speech detection on twitter. In *Proceedings of the NAACL Student Research Workshop*, pages 88–93, San Diego, California, June 2016. Association for Computational Linguistics.
- [3] Maarten Sap, Dallas Card, Saadia Gabriel, Yejin Choi, and Noah A. Smith. The risk of racial bias in hate speech detection. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 1668–1678, Florence, Italy, July 2019. Association for Computational Linguistics.
- [4] Joseph Fishkin. *Bottlenecks: A New Theory of Equal Opportunity*. Oxford University Press, New York, NY, 2014.
- [5] Shomik Jain, Vinith Suriyakumar, Kathleen Creel, and Ashia Wilson. Algorithmic pluralism: A structural approach to equal opportunity. In *The 2024 ACM Conference on Fairness, Accountability, and Transparency*, pages 197–206, Rio de Janeiro, Brazil, June 2024. ACM.
- [6] Shahadat Uddin, Arif Khan, Md Ekramul Hossain, and Mohammad Ali Moni. Comparing different supervised machine learning algorithms for disease prediction. *BMC Medical Informatics and Decision Making*, 19:281, 2019.
- [7] Schroders. Uk financial adviser survey 2025, 2025. Reports an increase in AI adoption among UK financial advisers from approximately 21% to 37% year-on-year.
- [8] Iginio Gagliardone, Danit Gal, Thiago Alves, and Gabriela Martinez. *Countering Online Hate Speech*. UNESCO Series on Internet Freedom. UNESCO Publishing, 2015.

- [9] Chris Bousquet. Limiting the right to moderate: Political equality, social media, and viewpoint-based moderation. *Ethics and Information Technology*, 27:62, 2025.
- [10] Manuel Tonneau, David Liu, Nikhil Malhotra, Scott A. Hale, Samuel P. Fraiberger, Victor Orozco-Olvera, and Paul Röttger. Hateday: Insights from a global hate speech dataset representative of a day on twitter, 2024.
- [11] Ifeoma Ajunwa. Automated employment discrimination. *Harvard Journal of Law and Technology*, 34:622–699, 2019.
- [12] David Schlangen. Targeting the benchmark: On methodology in current natural language processing research. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, pages 670–674, Aug 2021.
- [13] Richard Socher, Alex Perelygin, Jean Y. Wu, Jason Chuang, Christopher D. Manning, Andrew Y. Ng, and Christopher Potts. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1631–1642, Seattle, WA, 2013. Association for Computational Linguistics.
- [14] Bernard Koch, Emily Denton, Alex Hanna, and Jacob G. Foster. Reduced, reused and recycled: The life of a dataset in machine learning research. *arXiv preprint*, 2021.
- [15] Ondřej Bojar, Christian Buck, Christian Federmann, Barry Haddow, Philipp Koehn, Johannes Leveling, Christof Monz, Pavel Pecina, Matt Post, Herve Saint-Amand, Radu Soricut, Lucia Specia, and Aleš Tamchyna. Findings of the 2014 workshop on statistical machine translation. In *Proceedings of the Ninth Workshop on Statistical Machine Translation*, pages 12–58, Baltimore, Maryland, USA, June 2014. Association for Computational Linguistics.
- [16] Kwan Min Lee. Social media and political partisanship – a subaltern public sphere’s role in democracy. *Telematics and Informatics*, 35(7):1949–1957, 2018.
- [17] Mark Jurkowitz and Jeffrey Gottfried. Twitter is the go-to social media site for u.s. journalists, but not for the public. Pew Research Center Short Reads, June 2022. Accessed 2026-04-14.
- [18] Antigoni-Maria Founta, Constantinos Djouvas, Despoina Chatzakou, Ilias Leontiadis, Jeremy Blackburn, Gianluca Stringhini, Athena Vakali, Michael Sirivianos, and Nicolas Kourtellis. Large scale crowdsourcing and characterization of twitter abusive behavior. In *Proceedings of ICWSM*, 2018.
- [19] Despoina Chatzakou, Nicolas Kourtellis, Jeremy Blackburn, Emiliano De Cristofaro, Gianluca Stringhini, and Athena Vakali. Mean birds: Detecting aggression and bullying on twitter. In *Proceedings of the 9th ACM Web Science Conference*, 2017.

- [20] Jennifer Golbeck, Zahra Ashktorab, Rashad O. Banjo, Alexandra Berlinger, Sampada Bhagwan, Cody Buntain, Peter Cheakalos, Alan A. Geller, Quint Gregory, Ranjitha K. Gnanasekaran, Raghav R. Gunasekaran, Kimberly M. Hoffman, Jeremiah Hottle, Vishwa Jienjiltart, Shweta Khare, Ryan Lau, Mary J. Martindale, Sashank Naik, Heather L. Nixon, Pranav Ramachandran, Kristine M. Rogers, Lindsay Rogers, Ms. Shivani Sarin, Gunjan Shahane, Jay Thanki, Pooja Vengataraman, Zihe Wan, and Dajun M. Wu. A large labeled corpus for online harassment research. In *Proceedings of the 9th ACM Web Science Conference*, pages 229–233, 2017.
- [21] Albert-László Barabási and Réka Albert. Emergence of scaling in random networks. *Science*, 286(5439):509–512, 1999.
- [22] Florence E. Enock, Francesca Stevens, Jonathan Bright, Miranda Cross, Pica Johansson, Judy Wajcman, and Helen Z. Margetts. Understanding gender differences in experiences and concerns surrounding online harms: A short report on a nationally representative survey of uk adults. *CoRR*, abs/2402.00463, 2024.
- [23] Zeerak Waseem, James Thorne, and Joachim Bingel. Bridging the gaps: Multi task learning for domain transfer of hate speech detection. In *Proceedings of the 2nd Workshop on Abusive Language Online (ALW2)*, pages 29–40, Brussels, Belgium, 2018. Association for Computational Linguistics.
- [24] Su Lin Blodgett, Lisa Green, and Brendan O’Connor. Demographic dialectal variation in social media: A case study of african-american english. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 1119–1130, Austin, Texas, 2016. Association for Computational Linguistics.
- [25] Jacquelyn Rahman. The n word: Its history and use in the african american community. *Journal of English Linguistics*, 40:137–171, 2012.
- [26] Toon Calders, Faisal Kamiran, and Mykola Pechenizkiy. Building classifiers with independency constraints. In *2009 IEEE International Conference on Data Mining Workshops*, pages 13–18. IEEE, 2009.
- [27] Moritz Hardt, Eric Price, and Nati Srebro. Equality of opportunity in supervised learning. In *Proceedings of the 30th Conference on Neural Information Processing Systems (NeurIPS)*, 2016.
- [28] Ursula Hébert-Johnson, Michael Kim, Omer Reingold, and Guy Rothblum. Multicalibration: Calibration for the (computationally-identifiable) masses. In *Proceedings of the 35th International Conference on Machine Learning*, pages 1939–1948, Vienna, Austria, 2018. PMLR.
- [29] Joseph Fishkin and William E. Forbath. The anti-oligarchy constitution. *Boston University Law Review*, 94:669, 2014.
- [30] Bernard Williams. The idea of equality. In Peter Laslett and Walter Runciman, editors, *Philosophy, Politics and Society*, pages 112–117. Blackwell, Oxford, 1962.

- [31] Tammy Jiang, Jaimie L. Gradus, and Anthony J. Rosellini. Supervised machine learning: A brief primer. *Behavior Therapy*, 51(5):675–687, 2020.
- [32] Zayd Hammoudeh and Daniel Lowd. Training data influence analysis and estimation: A survey. *Machine Learning*, 2024.
- [33] X Corp. Dsa transparency report / content moderation systems documentation, 2024. Accessed 2026-04-14.
- [34] Digital Regulation Cooperation Forum. Auditing algorithms: The existing landscape, role of regulators and future outlook. Technical report, Digital Regulation Cooperation Forum, September 2022. Updated 23 September 2022.
- [35] Jon Kleinberg and Manish Raghavan. Algorithmic monoculture and social welfare. *Proceedings of the National Academy of Sciences*, 118(22):e2018340118, 2021.